Room Temperature Voltage Based Control of NV-Center Qubits

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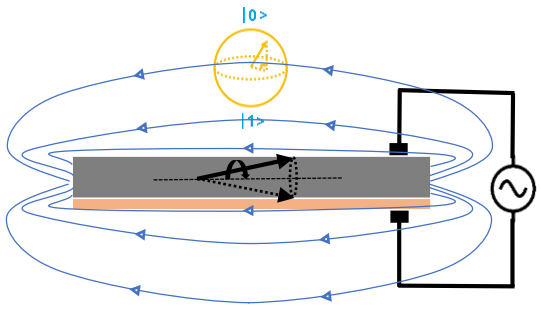
Approach

AC voltage across ferromagnets precesses its magnetic moment

Magnet exhibits an oscillating magnetic field

Microwaves

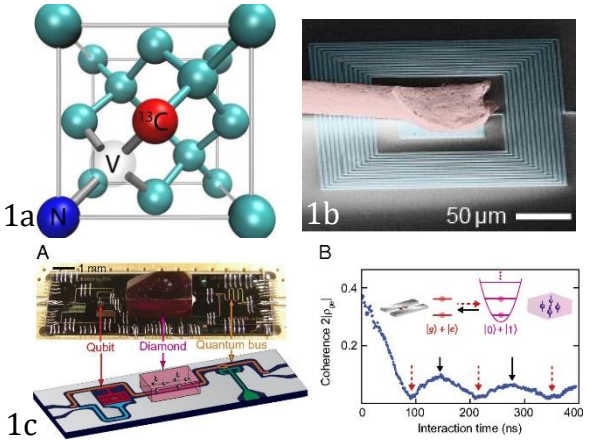
Nanomagnets of size 100x60 nm



(2) Setup for qubit control

Motivation

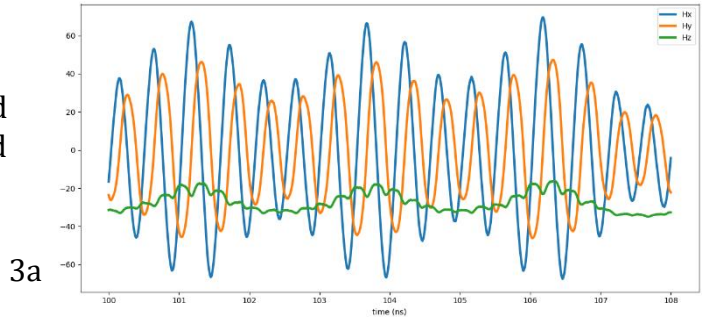
- Manipulation of NV-center spins at room temperature, to open possibilities for quantum technologies
- Need for a nanoscale microwave source

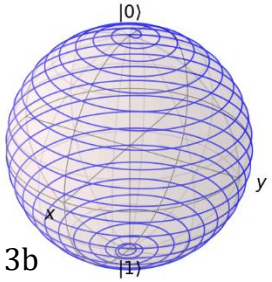


(1a) Nitrogen vacancy center in diamond
(1b) A typical on-chip microwave coil
(1c) Spin-ensemble quantum memory using NVC diamonds

Results

(3a) Simulated microwave field
(3b) Rabi oscillations exhibited by the qubit in presence of the microwaves on Bloch Sphere
(3c) Rabi oscillation in time domain





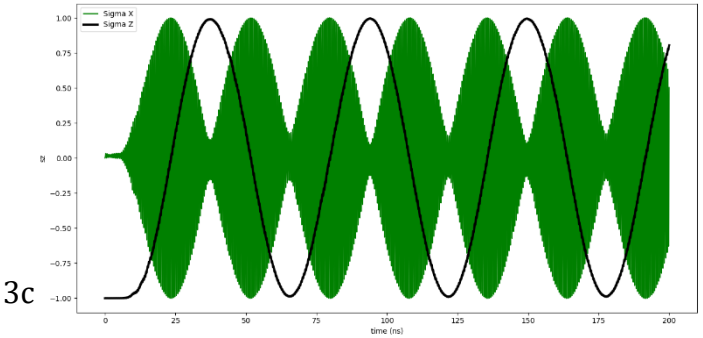


Image Sources:
1a: http://php.scripts.psu.edu/users/d/j/djs5809/
1b: Harabula et. al. Phys. Rev. Applied 8 (2017)
1c: http://www.pnas.org/content/112/13/3866

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